

# CensusInsight PK

## Pakistan Population Data Analysis System

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Generated on: 18 July 2026, 06:23 AM

### 1. Dataset Introduction

This report is based on the 7th Population & Housing Census 2023, conducted by the Pakistan Bureau of Statistics (PBS) — Pakistan's first fully digital census. The dataset covers all provinces and districts of Pakistan, including total population, urban/rural distribution, gender ratio, household counts, literacy and population density.

### 2. National Population Summary

| Metric                           | Value       |
|----------------------------------|-------------|
| Total National Population (2023) | 241,499,431 |
| Total Households                 | 38,248,926  |
| Urban Population %               | 48.1%       |
| Rural Population %               | 51.9%       |
| Number of Districts              | 136         |
| Number of Provinces              | 5           |

### 3. Population Density: Most & Least Crowded Districts

#### *Top 10 Most Densely Populated Districts*

| District   | Province           | Density (/km <sup>2</sup> ) |
|------------|--------------------|-----------------------------|
| Mardan     | Khyber Pakhtunkhwa | 12,054.9                    |
| Jhang      | Punjab             | 9,284.2                     |
| Peshawar   | Khyber Pakhtunkhwa | 4,933.9                     |
| Tharparkar | Sindh              | 4,532.1                     |
| Kech       | Balochistan        | 4,168.7                     |
| Keamari    | Sindh              | 3,698.9                     |
| Nowshera   | Khyber Pakhtunkhwa | 3,399.2                     |
| Swabi      | Khyber Pakhtunkhwa | 3,026.0                     |
| Haripur    | Khyber Pakhtunkhwa | 2,962.9                     |
| Khanewal   | Punjab             | 2,879.7                     |

#### *Top 10 Least Densely Populated Districts*

| District   | Province    | Density (/km <sup>2</sup> ) |
|------------|-------------|-----------------------------|
| Dera Bugti | Balochistan | 6.0                         |
| Jhal Magsi | Balochistan | 9.2                         |
| Nushki     | Balochistan | 16.5                        |
| Awaran     | Balochistan | 17.0                        |
| Kohlu      | Balochistan | 17.8                        |
| Loralai    | Balochistan | 18.5                        |
| Duki       | Balochistan | 19.2                        |
| Sibi       | Balochistan | 20.6                        |
| Kalat      | Balochistan | 23.1                        |
| Nasirabad  | Balochistan | 24.1                        |

#### 4. Urban vs Rural Breakdown (by Province)

| Province                    | Urban Population | Rural Population | Urban % |
|-----------------------------|------------------|------------------|---------|
| Balochistan                 | 6,006,223.0      | 8,888,179.0      | 40.3%   |
| Islamabad Capital Territory | 1,898,388.0      | 465,475.0        | 80.3%   |
| Khyber Pakhtunkhwa          | 14,791,334.0     | 26,064,763.0     | 36.2%   |
| Punjab                      | 65,751,653.0     | 61,937,269.0     | 51.5%   |
| Sindh                       | 27,775,512.0     | 27,920,635.0     | 49.9%   |

## 5. Key Gender Ratio Findings

The national gender ratio stands at **94.1 females per 100 males**. The district with the most balanced ratio and districts with notable imbalance are highlighted in the interactive dashboard (Module 5).

| District       | Province           | F per 100 M |
|----------------|--------------------|-------------|
| Upper Dir      | Khyber Pakhtunkhwa | 85.6        |
| Sargodha       | Punjab             | 86.0        |
| Duki           | Balochistan        | 87.9        |
| Jaffarabad     | Balochistan        | 88.1        |
| Lower Kohistan | Khyber Pakhtunkhwa | 88.3        |

## 6. Machine Learning Model Accuracy

| Model             | MAE    | R <sup>2</sup> Score |
|-------------------|--------|----------------------|
| Linear Regression | 46,347 | 0.9971               |
| Decision Tree     | 90,596 | 0.9839               |
| Random Forest     | 84,842 | 0.9819               |

Best performing model: **Linear Regression**

## 7. Sample District Population Projections

**Hangu, Khyber Pakhtunkhwa:** At the current growth rate of 1.9%, population is projected to reach **603,800** by 2030 and **663,708** by 2035.

**Bajaur, Khyber Pakhtunkhwa:** At the current growth rate of 2.4%, population is projected to reach **1,524,717** by 2030 and **1,720,035** by 2035.

**Nankana Sahib, Punjab:** At the current growth rate of 2.6%, population is projected to reach **1,963,343** by 2030 and **2,237,644** by 2035.

*Report auto-generated by CensusInsight PK. Data source: 7th Population & Housing Census 2023, Pakistan Bureau of Statistics (pbos.gov.pk).*